APPENDIX A

(REQUIRED FOR OPEN-BOOK EXAMINATION)

DESIGN DATA

BELT DRIVE

TABLES:

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Source: Uni-Drive Transmission Equipment – Roulunds Roflex V-belts

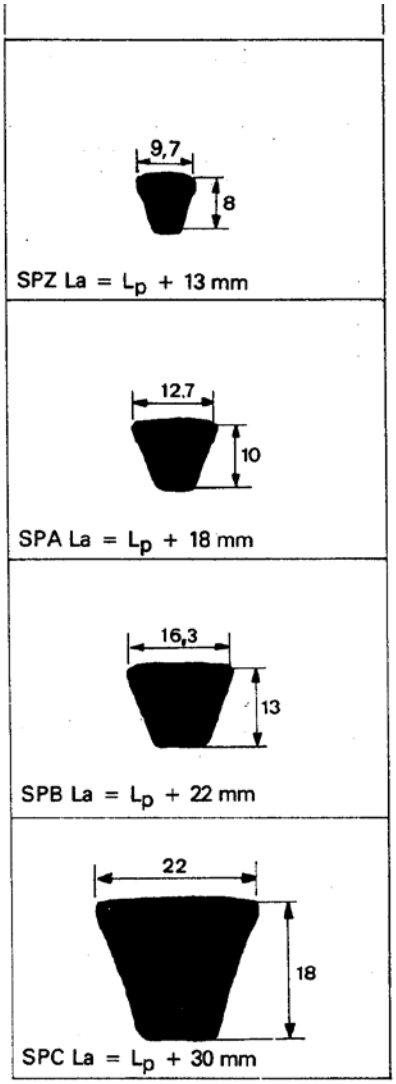
Table A-1: Cross Section Dimensions and Belt Lengths

Section	Dimension W x H	Recommended minimum diameter		Belt weight kg/m
SPZ	9,7 x 8	Pitch diameter d_p	63	0.070
SPA	12.7 x 10		90	0.119
SPB	16.3 x 13		140	0.194
SPC	22 x 18		224	0.360
19	18.6 x 15		180	0.250

Standard Pitch Length

SPZ	SPA	SPB	SPC
SPZ 630			
SPZ 670			
SPZ 710			
SPZ 750			
SPZ 800	SPA 800		
SPZ 850	SPA 850		
SPZ 900	SPA 900		
SPZ 950	SPA 950		
SPZ 1000	SPA 1000		
SPZ 1060	SPA 1060		
SPZ 1120	SPA 1120		
SPZ 1180	SPA 1180		
SPZ 1250	SPA 1250	**SPB 1250**	
SPZ 1320	SPA 1320	SPB 1320	
SPZ 1400	SPA 1400	**SPB 1400**	
SPZ 1500	SPA 1500	SPB 1500	
SPZ 1600	SPA 1600	**SPB 1600**	
SPZ 1700	SPA 1700	SPB 1700	
SPZ 1800	SPA 1800	**SPB 1800**	
SPZ 1900	SPA 1900	SPB 1900	
SPZ 2000	SPA 2000	**SPB 2000**	**SPC 2000**
SPZ 2120	SPA 2120	SPB 2120	SPC 2120
SPZ 2240	SPA 2240	**SPB 2240**	**SPC 2240**
SPZ 2360	SPA 2360	SPB 2360	SPC 2360
SPZ 2500	SPA 2500	**SPB 2500**	**SPC 2500**
SPZ 2650	SPA 2650	SPB 2650	SPC 2650
SPZ 2800	SPA 2800	**SPB 2800**	**SPC 2800**
SPZ 3000	SPA 3000	SPB 3000	SPC 3000
SPZ 3150	SPA 3150	**SPB 3150**	**SPC 3150**
SPZ 3350	SPA 3350	SPB 3350	SPC 3350
SPZ 3550	SPA 3550	**SPB 3550**	**SPC 3550**
	SPA 3750	SPB 3750	SPC 3750
	SPA 4000	**SPB 4000**	**SPC 4000**
	SPA 4250	SPB 4250	SPC 4250
	SPA 4500	**SPB 4500**	**SPC 4500**
		SPB 4750	SPC 4750
		SPB 5000	**SPC 5000**
		SPB 5300	SPC 5300
		SPB 5600	**SPC 5600**
		SPB 6000	SPC 6000
		SPB 6300	**SPC 6300**
		SPB 6700	SPC 6700
		SPB 7100	**SPC 7100**
		SPB 7500	SPC 7500
		SPB 8000	**SPC 8000**
		SPB 8500	SPC 8500
		SPB 9000	**SPC 9000**
			SPC 9500
			SPC 10000
			SPC 10600
			SPC 11200
			SPC 11800
			SPC 12500

(mm)



La = Outside length
L_p = Pitch length

Although the cross sections are identical, some manufacturers use other designations viz: —
SPZ is interchangeable with Alpha, 3V and 9N
SPB is interchangeable with Beta, 5V and 15N

Table A-2: **Service factors**

for different kinds of driver and driven machine combinations are shown below. The machines listed are representative examples only. When the particular machine considered is not listed, select a listed machine which has similar load characteristics.

DRIVEN UNIT	DRIVING UNIT					
	AC Motors, Normal Torque, Squirrel Cage, Synchronous and Split Phase. DC Motors, Shunt-Wound, Multiple Cylinder Internal Combustion Engines.			AC Motors, High Torque, High Slip, Repulsion – Induction, Single Phase, Series Wound and Slip Ring. DC Motors, Series Wound and Compound Wound. Single Cylinder Internal Combusion Engines. Line Shafts. Clutches.		
	Number of daily service hours			Number of daily service hours		
	<10	10-16	16-24	<10	10-16	16-24
Agitators for Liquids, Blowers and Exhausters, Fans up to 7.5 kW, Ligth Duty Conveyors	1,0	1,1	1,2	1,1	1,2	1,3
Belt conveyors for sand, grain, etc., Bough Mixers, Fans over 7.5 kW, Generators, Laundry Machinery, Machinery, Positive Displacement Rotary Pumps, Revolving and Vibrating Screens Line Shafts	1,1	1,2	1,3	1,2	1,3	1,4
Brick Machinery, Buckét Elevators, Piston Compressors, Conveyors (Drag-Pan-Screw), Hammer Mills, Paper Mill beaters, Piston Pumps, Positive Displacement Blowers, Pulverizers, Saw Mill and Woodworking Machinery, Textile Machinery	1,2	1,3	1,4	1,4	1,5	1,6
Crushers (Gyratory-Jaw-Roll), Mills (Ball-Rod), Moists, Rubber Calenders, Extruders – Mills.	1,3	1,4	1,5	1,5	1,6	1,8

Table A-3: **Choice of Cross Section**

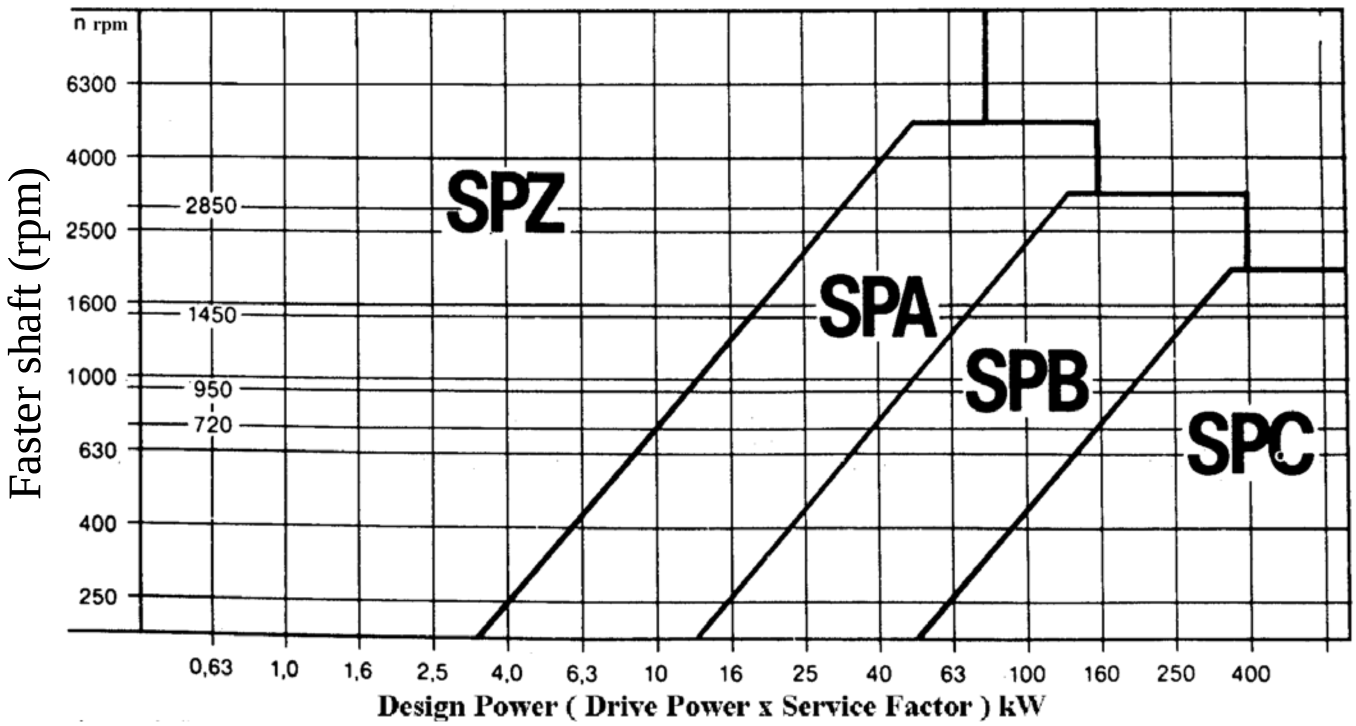


Table A-4: **Standard Sheave Diameters (mm)**

SPZ/3V	63	71	75	80	90	100	112	125	132	140	150	160	180	200	224	250	280	315	355	400	500	630			
SPA/4V	90	95	100	106	112	118	125	132	140	150	160	180	200	224	250	280	315	355	400	450	500	560	630	710	800
SPB/5V	140	150	160	170	180	200	224	250	280	315	355	400	450	500	560	600	710	750	800	900	1000	1120			
SPC/7V	224	236	250	265	280	300	315	335	355	400	450	500	560	600	630	710	750	800	900	1000	1120	1250	1400	1600	2000

Table A-5: **Correction Factor for Arc of Contact** C_θ

Arc of contact of smaller sheave	180°	174°	167°	163°	157°	151°	145°	139°	133°	127°	120°	113°	106°	99°	91°
C_θ	1.00	0.99	0.97	0.96	0.94	0.93	0.91	0.89	0.87	0.85	0.82	0.79	0.77	0.73	0.70

Table A-6: **Standard Belt Length L (mm) and Belt Length Correction Factor** C_L

Profile SPZ		Profile SPA		Profile SPB		Profile SPC	
L (mm)	Correction factor	L (mm)	Correction factor	L (mm)	Correction factor	L (mm)	Correction factor
630	0.85	800	0.81	1250	0.83	2000	0.83
710	0.87	900	0.83	1400	0.85	2240	0.85
800	0.89	1000	0.85	1600	0.87	2500	0.87
900	0.90	1120	0.87	1800	0.89	2800	0.89
1000	0.92	1250	0.89	2000	0.90	3150	0.90
1120	0.94	1400	0.90	2240	0.92	3550	0.92
1250	0.96	1600	0.93	2500	0.94	4000	0.94
1400	0.98	1800	0.94	2800	0.96	4500	0.96
1600	1.00	2000	0.96	3150	0.98	5000	0.98
1800	1.02	2240	0.98	3550	1.00	5600	1.00
2000	1.04	2500	1.00	4000	1.02	6300	1.02
2240	1.06	2800	1.02	4500	1.04	7100	1.04
2500	1.08	3150	1.04	5000	1.06	8000	1.06
2800	1.10	3550	1.06	5600	1.08	9000	1.08
3150	1.12	4000	1.08	6300	1.10	10000	1.10
3550	1.14	4500	1.10	7100	1.12	11200	1.12
				8000	1.14	12500	1.14

Standard belt lengths acc. to ISO 4148
Non-standard belt lengths available on request
Correction factor can be interpolated for intermediate belt lengths.

Table A-7a

kW-ratings per V-belt P [kW] at 180° arc of contact

SPZ

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave 200	400	800	1200	1600	2000	2400	3200	3600	4000	4500	5000	5500	6000
63	1.00	0.19	0.33	0.58	0.80	0.99	1.16	1.32	1.58	1.69	1.79	1.89	1.96	2.01	2.03
	1.05	0.19	0.35	0.61	0.84	1.04	1.23	1.39	1.69	1.81	1.92	2.03	2.12	2.19	2.23
	1.20	0.21	0.37	0.66	0.90	1.14	1.35	1.55	1.89	2.04	2.17	2.32	2.44	2.54	2.61
	1.50	0.22	0.39	0.70	0.97	1.22	1.45	1.67	2.05	2.22	2.37	2.54	2.69	2.81	2.91
	≥3.00	0.22	0.40	0.72	1.01	1.27	1.51	1.74	2.14	2.32	2.49	2.67	2.83	2.97	3.08
71	1.00	0.24	0.44	0.79	1.10	1.38	1.63	1.87	2.29	2.47	2.63	2.81	2.95	3.06	3.13
	1.05	0.25	0.46	0.81	1.13	1.43	1.70	1.95	2.39	2.59	2.76	2.95	3.11	3.24	3.33
	1.20	0.26	0.48	0.86	1.21	1.53	1.83	2.10	2.60	2.82	3.02	3.24	3.43	3.59	3.71
	1.50	0.27	0.50	0.90	1.27	1.61	1.93	2.22	2.76	3.00	3.22	3.47	3.68	3.86	4.01
	≥3.00	0.28	0.51	0.93	1.30	1.65	1.98	2.29	2.85	3.10	3.33	3.59	3.82	4.02	4.18
80	1.00	0.31	0.56	1.02	1.43	1.81	2.16	2.48	3.07	3.32	3.55	3.80	4.01	4.17	4.29
	1.05	0.31	0.58	1.04	1.47	1.86	2.22	2.56	3.17	3.44	3.68	3.95	4.17	4.35	4.48
	1.20	0.33	0.60	1.09	1.54	1.96	2.35	2.72	3.38	3.67	3.94	4.24	4.49	4.70	4.86
	1.50	0.34	0.62	1.13	1.60	2.04	2.45	2.84	3.54	3.85	4.14	4.46	4.74	4.98	5.16
	≥3.00	0.34	0.63	1.16	1.64	2.09	2.51	2.90	3.63	3.95	4.25	4.59	4.88	5.13	5.33
90	1.00	0.38	0.70	1.27	1.79	2.28	2.73	3.15	3.91	4.24	4.54	4.86	5.12	5.32	5.46
	1.05	0.39	0.71	1.30	1.83	2.33	2.80	3.23	4.01	4.36	4.67	5.00	5.29	5.50	5.66
	1.20	0.40	0.73	1.35	1.91	2.43	2.92	3.39	4.22	4.59	4.92	5.29	5.60	5.85	6.04
	1.50	0.41	0.76	1.39	1.97	2.51	3.02	3.51	4.38	4.77	5.12	5.52	5.85	6.13	6.34
	≥3.00	0.41	0.77	1.41	2.00	2.56	3.08	3.57	4.47	4.87	5.23	5.64	5.99	6.28	6.51
100	1.00	0.45	0.83	1.52	2.15	2.74	3.30	3.81	4.72	5.12	5.48	5.85	6.16	6.38	6.51
	1.05	0.45	0.84	1.55	2.19	2.80	3.36	3.89	4.83	5.24	5.61	6.00	6.32	6.56	6.70
	1.20	0.47	0.87	1.60	2.27	2.90	3.49	4.04	5.03	5.47	5.86	6.29	6.64	6.91	7.09
	1.50	0.48	0.89	1.64	2.33	2.98	3.59	4.16	5.19	5.65	6.06	6.51	6.89	7.18	7.39
	≥3.00	0.48	0.90	1.66	2.36	3.02	3.65	4.23	5.28	5.75	6.17	6.64	7.03	7.34	7.56
112	1.00	0.53	0.99	1.82	2.58	3.30	3.96	4.58	5.67	6.14	6.54	6.97	7.29	7.50	7.59
	1.05	0.54	1.00	1.84	2.62	3.35	4.03	4.66	5.77	6.25	6.68	7.12	7.45	7.68	7.78
	1.20	0.55	1.02	1.89	2.70	3.45	4.15	4.81	5.98	6.48	6.93	7.40	7.77	8.03	8.16
	1.50	0.56	1.04	1.93	2.76	3.53	4.25	4.93	6.14	6.66	7.13	7.63	8.02	8.30	8.47
	≥3.00	0.57	1.06	1.96	2.79	3.57	4.31	5.00	6.23	6.76	7.24	7.75	8.16	8.46	8.63
125	1.00	0.62	1.16	2.14	3.04	3.88	4.67	5.39	6.65	7.17	7.62	8.07	8.37	8.52	8.51
	1.05	0.63	1.17	2.16	3.08	3.93	4.73	5.47	6.75	7.29	7.75	8.21	8.54	8.70	8.71
	1.20	0.64	1.19	2.21	3.16	4.04	4.86	5.62	6.96	7.52	8.01	8.50	8.85	9.05	9.09
	1.50	0.65	1.21	2.25	3.22	4.12	4.96	5.74	7.12	7.70	8.21	8.72	9.10	9.33	9.39
	≥3.00	0.65	1.23	2.28	3.25	4.16	5.02	5.81	7.21	7.80	8.32	8.85	9.24	9.47	9.55
140	1.00	0.72	1.35	2.50	3.56	4.55	5.46	6.30	7.72	8.29	8.76	9.18	9.42	9.43	
	1.05	0.73	1.36	2.53	3.60	4.60	5.53	6.37	7.82	8.41	8.89	9.33	9.58	9.62	
	1.20	0.74	1.39	2.58	3.68	4.70	5.65	6.53	8.03	8.63	9.14	9.60	9.89	9.97	
	1.50	0.75	1.41	2.62	3.74	4.78	5.75	6.65	8.19	8.82	9.34	9.84	10.15	10.25	
	≥3.00	0.76	1.42	2.64	3.77	4.83	5.81	6.72	8.28	8.92	9.45	9.97	10.29	10.40	
160	1.00	0.86	1.60	2.98	4.24	5.41	6.48	7.45	9.03	9.62	10.06	10.39	10.43		
	1.05	0.86	1.62	3.01	4.28	5.47	6.55	7.53	9.14	9.74	10.20	10.53	10.59		
	1.20	0.88	1.64	3.06	4.36	5.57	6.68	7.68	9.34	9.97	10.45	10.82	10.91		
	1.50	0.89	1.66	3.10	4.42	5.65	6.78	7.80	9.50	10.15	10.65	11.05	11.16		
	≥3.00	0.89	1.67	3.12	4.45	5.69	6.83	7.87	9.59	10.25	10.76	11.17	11.30		
180	1.00	0.99	1.86	3.45	4.91	6.25	7.47	8.54	10.21	10.77	11.12	11.24			
	1.05	1.00	1.87	3.48	4.95	6.31	7.53	8.62	10.31	10.89	11.25	11.39			
	1.20	1.01	1.89	3.53	5.03	6.41	7.66	8.77	10.52	11.11	11.51	11.67			
	1.50	1.02	1.91	3.57	5.09	6.49	7.76	8.89	10.68	11.30	11.71	11.90			
	≥3.00	1.02	1.93	3.59	5.12	6.53	7.82	8.96	10.77	11.40	11.82	12.03			
200	1.00	1.12	2.11	3.92	5.57	7.07	8.41	9.55	11.24	11.71	11.90				
	1.05	1.13	2.12	3.94	5.61	7.12	8.47	9.64	11.34	11.82	12.03				
	1.20	1.14	2.15	3.99	5.69	7.22	8.60	9.79	11.55	12.05	12.28				
	1.50	1.15	2.17	4.03	5.75	7.30	8.70	9.91	11.71	12.23	12.48				
	≥3.00	1.16	2.18	4.06	5.78	7.35	8.76	9.97	11.80	12.34	12.59				

Dynamically balanced sheaves are requested.

Table A-7b

kW-ratings per V-belt P [kW] at 180° arc of contact **SPA**

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave 200	400	800	1200	1600	2000	2400	3200	3600	4000	4500	5000	5500	6000
90	1.00	0.42	0.74	1.30	1.77	2.18	2.54	2.85	3.32	3.48	3.57	3.61	3.55	3.38	3.10
	1.05	0.43	0.77	1.36	1.86	2.30	2.70	3.04	3.57	3.75	3.88	3.96	3.94	3.81	3.57
	1.20	0.46	0.83	1.48	2.04	2.54	3.00	3.40	4.05	4.29	4.48	4.63	4.69	4.63	4.46
	1.50	0.49	0.88	1.57	2.18	2.73	3.23	3.68	4.43	4.72	4.96	5.17	5.28	5.28	5.17
	≥3.00	0.50	0.90	1.62	2.26	2.84	3.36	3.84	4.64	4.96	5.22	5.47	5.61	5.65	5.57
100	1.00	0.53	0.96	1.71	2.37	2.96	3.49	3.95	4.69	4.96	5.15	5.28	5.28	5.13	4.84
	1.05	0.55	0.99	1.77	2.47	3.09	3.64	4.14	4.94	5.24	5.46	5.63	5.67	5.56	5.30
	1.20	0.58	1.05	1.89	2.64	3.32	3.94	4.50	5.42	5.77	6.06	6.30	6.41	6.38	6.20
	1.50	0.60	1.10	1.99	2.79	3.51	4.18	4.78	5.80	6.20	6.53	6.83	7.00	7.03	6.91
	≥3.00	0.62	1.13	2.04	2.87	3.62	4.31	4.94	6.01	6.44	6.80	7.13	7.34	7.40	7.31
112	1.00	0.67	1.23	2.21	3.09	3.89	4.60	5.24	6.27	6.66	6.94	7.14	7.16	6.99	6.61
	1.05	0.69	1.26	2.27	3.18	4.01	4.76	5.43	6.52	6.93	7.25	7.49	7.55	7.41	7.07
	1.20	0.72	1.32	2.39	3.36	4.25	5.06	5.79	7.00	7.47	7.84	8.16	8.30	8.24	7.97
	1.50	0.74	1.36	2.49	3.50	4.44	5.29	6.07	7.38	7.90	8.32	8.68	8.89	8.89	8.68
	≥3.00	0.75	1.39	2.54	3.58	4.54	5.43	6.23	7.59	8.14	8.58	8.99	9.22	9.25	9.07
125	1.00	0.82	1.51	2.74	3.86	4.87	5.79	6.60	7.91	8.39	8.72	8.96	8.95	8.67	8.10
	1.05	0.83	1.54	2.81	3.95	4.99	5.94	6.79	8.16	8.67	9.04	9.30	9.33	9.09	8.55
	1.20	0.86	1.60	2.92	4.13	5.23	6.24	7.15	8.64	9.21	9.64	9.97	10.08	9.91	9.46
	1.50	0.89	1.65	3.02	4.27	5.42	6.48	7.43	9.02	9.63	10.12	10.52	10.67	10.56	10.17
	≥3.00	0.90	1.67	3.07	4.35	5.53	6.61	7.59	9.22	9.87	10.38	10.81	11.00	10.93	10.57
140	1.00	0.99	1.83	3.35	4.73	5.99	7.12	8.12	9.70	10.25	10.62	10.80	10.65	10.13	
	1.05	1.00	1.86	3.41	4.82	6.11	7.27	8.31	9.95	10.53	10.93	11.15	11.04	10.56	
	1.20	1.03	1.92	3.53	5.00	6.35	7.57	8.67	10.43	11.07	11.52	11.82	11.78	11.38	
	1.50	1.06	1.97	3.63	5.15	6.54	7.81	8.95	10.81	11.49	12.00	12.35	12.38	12.03	
	≥3.00	1.07	1.99	3.68	5.22	6.64	7.94	9.10	11.02	11.73	12.26	12.65	12.71	12.40	
160	1.00	1.21	2.26	4.15	5.88	7.44	8.84	10.06	11.89	12.46	12.78	12.76	12.26		
	1.05	1.23	2.29	4.22	5.97	7.56	8.99	10.24	12.14	12.74	13.08	13.11	12.65		
	1.20	1.26	2.35	4.34	6.15	7.80	9.29	10.60	12.62	13.28	13.68	13.78	13.39		
	1.50	1.28	2.39	4.43	6.29	7.99	9.53	10.89	13.00	13.71	14.16	14.32	13.99		
	≥3.00	1.29	2.42	4.48	6.37	8.10	9.66	11.05	13.21	13.95	14.42	14.61	14.32		
180	1.00	1.43	2.68	4.95	7.00	8.85	10.49	11.88	13.85	14.35	14.49	14.10			
	1.05	1.45	2.71	5.01	7.09	8.97	10.64	12.07	14.09	14.63	14.80	14.45			
	1.20	1.48	2.77	5.13	7.27	9.21	10.94	12.43	14.57	15.17	15.40	15.13			
	1.50	1.50	2.82	5.22	7.41	9.39	11.18	12.71	14.95	15.59	15.87	15.66			
	≥3.00	1.52	2.84	5.27	7.49	9.51	11.31	12.87	15.16	15.83	16.14	15.96			
200	1.00	1.65	3.10	5.73	8.10	10.22	12.06	13.59	15.54	15.87	15.72				
	1.05	1.67	3.13	5.79	8.19	10.34	12.22	13.78	15.79	16.15	16.03				
	1.20	1.70	3.19	5.91	8.37	10.58	12.52	14.13	16.27	16.69	16.62				
	1.50	1.72	3.23	6.00	8.51	10.77	12.75	14.42	16.64	17.12	17.10				
	≥3.00	1.74	3.26	6.06	8.59	10.88	12.88	14.58	16.86	17.35	17.36				
224	1.00	1.92	3.59	6.65	9.39	11.80	13.85	15.47	17.20	17.17					
	1.05	1.93	3.62	6.71	9.47	11.93	14.00	15.65	17.44	17.45					
	1.20	1.96	3.68	6.83	9.66	12.17	14.30	16.01	17.92	17.99					
	1.50	1.99	3.73	6.93	9.80	12.36	14.54	16.30	18.30	18.42					
	≥3.00	2.00	3.76	6.98	9.88	12.46	14.67	16.46	18.51	18.66					
250	1.00	2.20	4.13	7.64	10.75	13.44	15.65	17.28	18.48						
	1.05	2.21	4.16	7.70	10.84	13.57	15.80	17.46	18.73						
	1.20	2.24	4.22	7.82	11.02	13.81	16.10	17.82	19.21						
	1.50	2.27	4.26	7.91	11.17	14.00	16.34	18.11	19.59						
	≥3.00	2.28	4.29	7.96	11.25	14.10	16.47	18.27	19.80						
280	1.00	2.52	4.73	8.75	12.27	15.23	17.53	19.05	19.24						
	1.05	2.54	4.77	8.80	12.36	15.35	17.68	19.23	19.49						
	1.20	2.57	4.83	8.93	12.54	15.59	17.98	19.59	19.96						
	1.50	2.59	4.87	9.03	12.68	15.78	18.22	19.87	20.34						
	≥3.00	2.60	4.90	9.08	12.76	15.89	18.35	20.03	20.56						
315	1.00	2.89	5.44	10.02	13.97	17.16	19.44	20.63							
	1.05	2.91	5.47	10.09	14.06	17.28	19.59	20.82							
	1.20	2.94	5.53	10.21	14.24	17.52	19.89	21.18							
	1.50	2.96	5.57	10.30	14.38	17.71	20.13	21.46							
	≥3.00	2.98	5.60	10.35	14.46	17.82	20.26	21.62							

Dynamically balanced sheaves are requested.

Table A-7c

kW-ratings per V-belt P [kW] at 180° arc of contact — SPB

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave: 200	400	800	1200	1600	1800	2000	2200	2400	3200	3600	4000	4500	5000
140	1.00	1.20	2.17	3.85	5.30	6.55	7.10	7.60	8.05	8.45	9.43	9.54	9.35	8.68	7.45
	1.05	1.24	2.24	3.99	5.50	6.82	7.41	7.94	8.43	8.85	9.97	10.15	10.04	9.43	8.30
	1.20	1.30	2.37	4.25	5.90	7.35	8.00	8.60	9.14	9.64	11.03	11.34	11.35	10.92	9.93
	1.50	1.35	2.47	4.46	6.21	7.76	8.47	9.12	9.72	10.27	11.87	12.27	12.39	12.09	11.24
	≥3.00	1.38	2.53	4.57	6.38	8.00	8.72	9.41	10.04	10.62	12.33	12.80	12.97	12.74	11.97
160	1.00	1.57	2.86	5.15	7.16	8.91	9.68	10.40	11.03	11.59	13.01	13.15	12.87	11.87	10.08
	1.05	1.60	2.93	5.29	7.36	9.18	9.99	10.74	11.41	12.00	13.55	13.76	13.55	12.64	10.93
	1.20	1.67	3.06	5.55	7.76	9.71	10.59	11.39	12.13	12.79	14.60	14.95	14.87	14.12	12.57
	1.50	1.72	3.17	5.76	8.07	10.12	11.05	11.91	12.70	13.41	15.43	15.88	15.91	15.29	13.87
	≥3.00	1.75	3.22	5.88	8.25	10.36	11.32	12.21	13.02	13.76	15.90	16.41	16.49	15.94	14.60
180	1.00	1.93	3.55	6.44	8.99	11.21	12.19	13.08	13.87	14.56	16.19	16.22	15.67	14.07	
	1.05	1.96	3.61	6.57	9.18	11.48	12.49	13.42	14.25	14.97	16.73	16.83	16.35	14.83	
	1.20	2.03	3.75	6.84	9.58	12.00	13.09	14.08	14.97	15.76	17.78	18.02	17.66	16.31	
	1.50	2.08	3.85	7.05	9.89	12.42	13.55	14.60	15.54	16.38	18.61	18.95	18.70	17.48	
	≥3.00	2.11	3.91	7.16	10.07	12.65	13.82	14.89	15.86	16.73	19.08	19.48	19.29	18.14	
200	1.00	2.28	4.23	7.71	10.78	13.43	14.60	15.64	16.56	17.34	18.95	18.71	17.67		
	1.05	2.32	4.29	7.84	10.98	13.71	14.90	15.98	16.93	17.75	19.49	19.32	18.35		
	1.20	2.38	4.42	8.11	11.37	14.23	15.49	16.64	17.66	18.54	20.54	20.50	19.67		
	1.50	2.43	4.53	8.31	11.69	14.65	15.96	17.16	18.23	19.16	21.37	21.44	20.71		
	≥3.00	2.46	4.59	8.43	11.86	14.88	16.23	17.45	18.55	19.51	21.84	21.96	21.29		
224	1.00	2.71	5.03	9.21	12.87	16.01	17.36	18.55	19.57	20.40	21.65	20.83			
	1.05	2.74	5.10	9.34	13.08	16.28	17.67	18.89	19.94	20.81	22.19	21.44			
	1.20	2.81	5.23	9.60	13.47	16.81	18.26	19.55	20.66	21.60	23.24	22.63			
	1.50	2.86	5.34	9.82	13.78	17.22	18.73	20.07	21.24	22.22	24.07	23.56			
	≥3.00	2.89	5.39	9.93	13.96	17.46	18.99	20.36	21.56	22.57	24.54	24.09			
250	1.00	3.17	5.90	10.81	15.09	18.68	20.18	21.48	22.54	23.35	23.75				
	1.05	3.20	5.97	10.95	15.29	18.95	20.49	21.82	22.91	23.76	24.29				
	1.20	3.27	6.10	11.21	15.69	19.48	21.08	22.47	23.63	24.54	25.34				
	1.50	3.32	6.20	11.42	16.00	19.89	21.55	22.99	24.20	25.17	26.17				
	≥3.00	3.35	6.26	11.53	16.17	20.13	21.81	23.28	24.53	25.52	26.64				
280	1.00	3.69	6.89	12.62	17.56	21.59	23.21	24.54	25.56	26.23	24.99				
	1.05	3.72	6.96	12.76	17.76	21.86	23.52	24.88	25.93	26.64	25.53				
	1.20	3.79	7.09	13.02	18.16	22.38	24.11	25.54	26.65	27.43	26.59				
	1.50	3.84	7.19	13.23	18.47	22.80	24.57	26.06	27.22	28.05	27.42				
	≥3.00	3.87	7.25	13.35	18.64	23.03	24.84	26.35	27.55	28.40	27.88				
315	1.00	4.29	8.03	14.69	20.32	24.73	26.39	27.66	28.48	28.82					
	1.05	4.33	8.10	14.83	20.53	25.00	26.70	28.00	28.85	29.23					
	1.20	4.39	8.23	15.09	20.92	25.53	27.29	28.65	29.57	30.02					
	1.50	4.45	8.33	15.30	21.23	25.94	27.76	29.17	30.15	30.64					
	≥3.00	4.48	8.39	15.42	21.41	26.17	28.02	29.46	30.47	30.99					
355	1.00	4.98	9.32	17.00	23.31	27.96	29.54	30.55	30.94						
	1.05	5.01	9.38	17.13	23.52	28.23	29.84	30.89	31.31						
	1.20	5.08	9.52	17.40	23.91	28.76	30.43	31.54	32.04						
	1.50	5.13	9.62	17.60	24.22	29.17	30.90	32.06	32.61						
	≥3.00	5.16	9.68	17.72	24.40	29.41	31.16	32.36	32.93						
400	1.00	5.74	10.74	19.51	26.45	31.09	32.37	32.86							
	1.05	5.78	10.81	19.64	26.66	31.37	32.68	33.20							
	1.20	5.84	10.94	19.91	27.05	31.89	33.27	33.85							
	1.50	5.90	11.05	20.11	27.36	32.31	33.74	34.37							
	≥3.00	5.92	11.11	20.23	27.54	32.54	34.00	34.67							

Dynamically balanced sheaves are requested.

Table A-7d

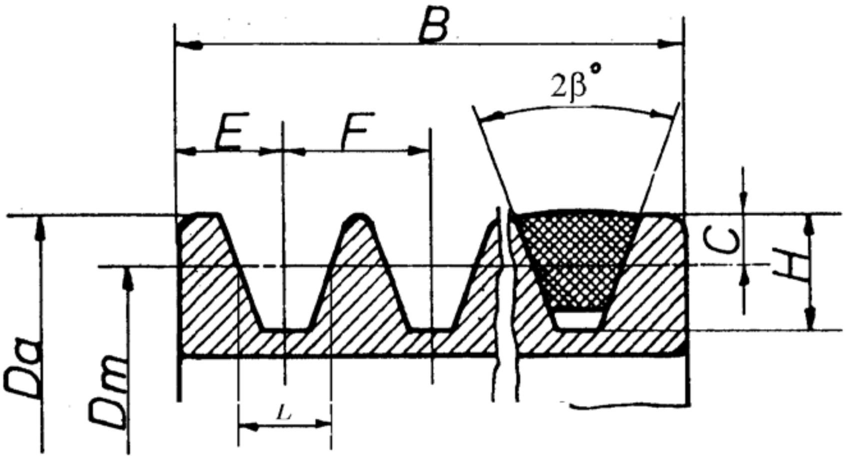
kW-ratings per V-belt P [kW] at 180° arc of contact **SPC**

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave 200	300	400	500	600	800	1200	1600	1800	2000	2200	2400	2600	3200
224	1.00	3.63	5.15	6.58	7.94	9.22	11.64	15.77	18.94	20.11	20.99	21.56	21.78	21.64	18.78
	1.05	3.72	5.29	6.77	8.17	9.51	12.01	16.33	19.68	20.95	21.92	22.58	22.89	22.84	20.26
	1.20	3.90	5.56	7.13	8.62	10.05	12.73	17.41	21.12	22.57	23.72	24.56	25.05	25.18	23.14
	1.50	4.05	5.77	7.41	8.97	10.48	13.30	18.27	22.26	23.85	25.15	26.13	26.76	27.03	25.42
	≥3.00	4.13	5.89	7.57	9.18	10.72	13.62	18.74	22.90	24.57	25.95	27.00	27.72	28.07	26.70
250	1.00	4.48	6.38	8.19	9.89	11.54	14.61	19.89	23.90	25.37	26.44	27.08	27.27	26.95	22.68
	1.05	4.57	6.52	8.37	10.14	11.82	14.98	20.44	24.64	26.21	27.37	28.11	28.38	28.16	24.17
	1.20	4.75	6.79	8.72	10.59	12.36	15.70	21.52	26.08	27.83	29.17	30.09	30.54	30.50	27.05
	1.50	4.89	7.00	9.02	10.94	12.79	16.27	22.38	27.22	29.11	30.59	31.65	32.25	32.35	29.33
	≥3.00	4.97	7.12	9.18	11.14	13.03	16.59	22.86	27.86	29.83	31.39	32.53	33.21	33.39	30.60
280	1.00	5.44	7.79	10.02	12.14	14.18	17.98	24.48	29.31	31.00	32.15	32.71	32.63	31.87	
	1.05	5.54	7.93	10.20	12.38	14.46	18.35	25.04	30.05	31.84	33.08	33.73	33.75	33.08	
	1.20	5.72	8.20	10.56	12.83	15.00	19.07	26.12	31.49	33.46	34.88	35.71	35.91	35.42	
	1.50	5.86	8.41	10.85	13.18	15.42	19.64	26.97	32.63	34.74	36.30	37.28	37.61	37.27	
	≥3.00	5.94	8.53	11.01	13.38	15.66	19.96	27.45	33.27	35.46	37.10	38.15	38.57	38.31	
315	1.00	6.56	9.42	12.13	14.72	17.20	21.83	29.62	35.16	36.93	37.95	38.15	37.45	35.80	
	1.05	6.66	9.55	12.32	14.96	17.48	22.20	30.18	35.90	37.76	38.88	39.17	38.57	37.01	
	1.20	6.84	9.83	12.68	15.41	18.02	22.92	31.26	37.34	39.38	40.68	41.15	40.73	39.35	
	1.50	6.98	10.04	12.96	15.76	18.45	23.49	32.11	38.48	40.67	42.10	42.71	42.44	41.20	
	≥3.00	7.06	10.16	13.12	15.96	18.69	23.81	32.59	39.12	41.38	42.90	43.59	43.40	42.24	
355	1.00	7.83	11.26	14.52	17.63	20.60	26.11	35.19	41.17	42.78	43.33	42.73			
	1.05	7.92	11.40	14.71	17.86	20.88	26.48	35.75	41.92	43.62	44.26	43.75			
	1.20	8.10	11.67	15.07	18.31	21.42	27.20	36.83	43.36	45.23	46.06	45.73			
	1.50	8.25	11.88	15.35	18.67	21.85	27.77	37.68	44.50	46.52	47.48	47.29			
	≥3.00	8.33	12.00	15.51	18.87	22.09	28.09	38.16	45.13	47.23	48.28	48.17			
400	1.00	9.24	13.31	17.17	20.85	24.35	30.77	41.03	47.01	48.06	47.63				
	1.05	9.34	13.45	17.36	21.08	24.62	31.15	41.59	47.75	48.90	48.56				
	1.20	9.52	13.72	17.72	21.53	25.16	31.87	42.67	49.19	50.52	50.36				
	1.50	9.66	13.93	18.00	21.89	25.59	32.44	43.52	50.33	51.80	51.78				
	≥3.00	9.74	14.05	18.16	22.09	25.83	32.75	44.00	50.97	52.52	52.58				
450	1.00	10.80	15.56	20.07	24.35	28.41	35.76	46.95	52.22	52.14					
	1.05	10.89	15.70	20.26	24.59	28.69	36.13	47.51	52.96	52.97					
	1.20	11.07	15.97	20.62	25.04	29.23	36.85	48.59	54.40	54.59					
	1.50	11.22	16.18	20.90	25.39	29.65	37.42	49.44	55.54	55.87					
	≥3.00	11.30	16.30	21.06	25.59	29.89	37.74	49.92	56.18	56.59					
500	1.00	12.34	17.78	22.93	27.79	32.36	40.53	52.21	55.94						
	1.05	12.44	17.92	23.11	28.02	32.64	40.90	52.77	56.68						
	1.20	12.62	18.19	23.47	28.47	33.18	41.62	53.85	58.12						
	1.50	12.76	18.40	23.76	28.83	33.60	42.19	54.70	59.26						
	≥3.00	12.84	18.52	23.92	29.03	33.84	42.51	55.18	59.90						
560	1.00	14.17	20.41	26.29	31.82	36.95	45.94	57.58	58.25						
	1.05	14.27	20.55	26.48	32.05	37.23	46.31	58.13	58.99						
	1.20	14.45	20.82	26.84	32.50	37.77	47.03	59.21	60.43						
	1.50	14.59	21.04	27.13	32.85	38.20	47.60	60.07	61.57						
	≥3.00	14.67	21.16	27.29	33.05	38.44	47.92	60.55	62.21						
630	1.00	16.29	23.44	30.14	36.37	42.09	51.79	62.40							
	1.05	16.38	23.58	30.33	36.60	42.37	52.16	62.96							
	1.20	16.56	23.85	30.69	37.05	42.91	52.88	64.04							
	1.50	16.70	24.06	30.97	37.41	43.34	53.45	64.89							
	≥3.00	16.78	24.18	31.13	37.61	43.58	53.77	65.37							
710	1.00	18.68	26.84	34.42	41.39	47.66	57.80	65.81							
	1.05	18.77	26.98	34.61	41.62	47.94	58.17	66.37							
	1.20	18.95	27.25	34.97	42.07	48.48	58.89	67.45							
	1.50	19.09	27.46	35.25	42.42	48.91	59.46	68.30							
	≥3.00	19.17	27.58	35.41	42.62	49.15	59.78	68.78							

Dynamically balanced sheaves are requested.

Table A-8

TAPER LOCK PULLEY



The pulleys conform to
ISO 4183 – 1980 and DIN 2211
Designed for use with both
Wedge (narrow) and
classical V-belts

B = F (n – 1) + 2 E

n = No. of Grooves
Dimensions in millimetres

Profile	Dm	2β	L	E	F	C	H	Effective Coff Friction, fe
SPZ	≦ 80 > 80	34 38	8,5	8	12	2,00	11	0.45 0.40
SPA	≦ 118 > 118	34 38	11	10	15	2,75	13,75	0.45 0.40
SPB	≦ 190 > 190	34 38	14	12,5	19	3,50	17,50	0.45 0.40
SPC	≦ 315 > 315	34 38	19	17	25,5	4,8	24	0.45 0.40

Material:
Fine grain cast-iron – Grade 260–200

Eccentricity of O/D to bore:
DIN 2211 part 1

Groove side wobble tolerances:
DIN 2211 part 1

Surface roughness:
ISO 254 – 1981

Balance quality grade:
ISO 1940 – 1973
Pulleys with dm (pcd) > 125 mm are static balanced to grade q 6,3.
A residual unbalance of 2 g on the pulley dm (pcd) will be accepted.
The residual unbalance is ≦ 100 g mm/kg at 600 rev./min. and 20 g mm/kg at 3000 rev./min.
Dynamic balancing can be supplied to order.